AMERICAN ACADEMY OF OPHTHALMOLOGY®

Ultrasonographic Assessment of Intrinsic Vascularity in Choroidal Nevus, Suspicious Choroidal Nevus, and Choroidal Melanoma

Buse Guneri Beser, MD, Tanya McClendon, CDOS, Bernadete Ayres, MD, Hakan Demirci, MD

Purpose: To determine if intrinsic vascularity can be observed in B-scan ultrasound examinations of choroidal nevi (CN) and compare the intensity of the vascularity among CN, suspicious choroidal nevi (SN), and choroidal melanomas (CMs).

Design: A prospective nonrandomized comparative trial.

Participants: Patients with CN, SN, or treatment-naive CM.

Methods: From September 2016 to January 2023, all patients underwent ultrasound examination using 10- or 15-MHz B-scan probes at a tertiary ocular oncology service.

Main Outcome Measures: Observed intrinsic vascularity on B-scan examinations.

Results: Forty-three eyes with CN, 49 with SN, and 45 with CM were included. We evaluated the areas of flickering motion inside the lesions on B-scan ultrasonography (USG) videos. Intrinsic vascularity was detected in 37 (86%) CN, 43 (88%) SN, and 45 (100%) CM. The intensity of intrinsic vascularity was moderate in 58% of CN but high in 45% of SN and 73% of melanomas. In the combined group (nevus and melanoma), the degree of intensity of intrinsic vascularity demonstrated a positive correlation with subretinal fluid and the lesions` height and basal diameter. In contrast, a negative correlation was observed between the degree of intensity of intrinsic vascularity and internal reflectivity and the presence of overlying drusen.

Conclusions: Choroidal nevi can have moderate intrinsic vascularity, detectable during B-scan ultrasound examinations. This contradicts the current view that CN is a nonvascular tumor on ophthalmic ultrasound. Therefore, the identification of vascular motion in B-scan USG cannot be used to exclude the diagnosis of CN. Serial multimodal imaging along with a clinical follow-up is necessary for the differentiation of CN and melanomas.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science 2025;5:100822* *© 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Supplemental material available at www.ophthalmologyscience.org.

Choroidal nevus (CN) is the most common benign intraocular tumor.[1] In the United States, the prevalence of CN in adults above 49 years of age is 4.7%.[2] Choroidal nevi are typically flat or minimally elevated, pigmented, and well-circumscribed lesions.[3] There is no strict cutoff value regarding dimensions or shape to differentiate them from choroidal melanoma (CM). In a study that comprised a medical record review of 4100 patients diagnosed with CN at a tertiary ocular oncology center, the mean basal diameter and thickness of CN were 5 mm and 1.5 mm, respectively.[4]

Differentiating between benign CN and small CMs can be challenging. Choroidal nevi and CM can demonstrate similar clinical features, including tumor size, color (pigmented or nonpigmented), location, overlying retinal pigment epithelium (RPE) alterations or orange pigment, and associated subretinal fluid (SRF). The implementation of serial multimodal imaging techniques, which includes ultra-widefield fundus photography, fundus autofluorescence (FAF), enhanced-depth imaging OCT, A- and B-scan ultrasonography (USG), and various angiography methods such as fundus fluorescein angiography and, more recently, OCT angiography, along with clinical follow-up, is important in differentiating CN from CM.[5] This comprehensive strategy is essential for identifying changes in lesion size and associated risk factors, ultimately contributing to improved patient care and outcomes.

Most CN demonstrate high internal reflectivity on A-scan USG, whereas melanomas may show low-to-medium internal reflectivity.[6] As small melanomas grow, their internal reflectivity may decrease to low-to-medium. Classically, CN are defined as nonvascular echographic lesions, while melanomas may show sound attenuation, excavation of choroidal tissue, orbital shadowing, and spontaneous vascular pulsations on ocular USG.[7]

In this study, we evaluated if the intrinsic vascularity is observed in CN on B-scan USG examinations when

© 2025 by the American Academy of Ophthalmology
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/). Published by Elsevier Inc.

https://doi.org/10.1016/j.xops.2025.100822
ISSN 2666-9145/25

performed with conventional (10-MHz or 15-MHz) probes. Additionally, we compared the intensity of the intrinsic vascularity of the CN, suspicious choroidal nevus (SN), and CM and assessed correlations between lesion clinical features and the degree of intensity of intrinsic vascularity.

Methods

The study adhered to the tenets of the Declaration of Helsinki. The patients underwent ultrasonographic examination as a part of the standard of care. The review of the medical records and ultrasonographic examination was conducted under the study approved by the Institutional Review Board of the University of Michigan. Therefore, the informed consent for each patient was not applicable. Consecutive treatment-naive unilateral unifocal CN or consecutive treatment-naive CMs undergoing the ultrasonographic examination between September 2016 and January 2023 were included.

Patients were prospectively identified and received a complete dilated eye examination with fundus drawing, USG (Aviso Platform, Quantel Medical), ultra-widefield fundus photography (Optos, Inc), FAF (Optos, Inc), and enhanced-depth imaging OCT of the macula and the retina overlying the lesion (Heidelberg Engineering). Fundus fluorescein angiography (Optos, Inc) was obtained as needed in CN cases associated with choroidal neovascular membrane. For amelanotic lesions, indocyanine green angiography (Optos, Inc) was performed to rule out choroidal hemangioma, and systemic workup with computerized tomography of the chest, abdomen, and pelvis or mammography was ordered to rule out metastasis. Lesion clinical features determined by dilated fundoscopy examinations performed by a single ocular oncologist (H.D.) were recorded and used as the basis for diagnosis.

This study included 43 cases of CN measuring less than 2.5 mm in thickness, lacking high-risk features for tumor growth such as SRF on OCT or orange pigment on FAF but exhibiting overlying drusen. In contrast, lesions $\geq$3.0 mm in thickness (41 cases) were classified as CM. Lesions that are $<$3.0 mm in thickness and either have associated SRF or orange pigment were categorized into the SN group, comprising 53 cases. Among the SN group, 16 patients chose to undergo plaque radiotherapy after the initial presentation, 4 patients opted for a diagnostic fine needle aspiration biopsy (FNAB), and 33 patients were observed carefully for at least 2 years. This study included the results of FNAB for assessment of gene expression profile (GEP) with DecisionDx-UM (class 1A, class 1B, or class 2) and cancer testis antigen preferentially expressed antigen in melanoma (PRAME) RNA expression status with Decision-Dx PRAME (negative or positive) to assess the metastatic risk. This testing was performed by Castle Biosciences. During the 2-year follow-up period, 23 patients with SN showed no growth, while 10 patients exhibited documented growth. From the 4 patients who opted for an FNAB for diagnostic purposes, 2 showed cytology consistent with CM, while the other 2 had a class 1B GEP and insufficient cellularity on cytology. These 4 cases were included in the CM group. Ultimately, the study included 49 eyes with SN and 45 eyes with CM. Lesions measuring less than 1.2 mm in thickness by USG were excluded from the study because the intrinsic vascularity could not be evaluated in these lesions.

The detailed ocular history, follow-up, and treatment processes were obtained from electronic medical health records. Demographic data included patient age, sex, and eye laterality. Overlying RPE changes or drusen (negative or positive), orange pigment (negative or positive), and associated SRF (negative or positive) were assessed from fundus pictures, FAF, and OCT, respectively.

The ocular ultrasonographic data were collected as listed below.

Ocular USG Acquisition Protocol

All ultrasonographic examinations were performed by 3 experienced sonographers using the direct contact technique under topical anesthesia (proparacaine hydrochloride ophthalmic solution, Alcaine 0.5%, Alcon-USA) over an open eye after the placement of a lubricant gel (carboxymethylcell-hypromellose, Genteal Gel 0.25%–0.3% Alcon-USA). In B-mode USG, the gain was adjusted for each probe to produce optimal images of the choroidal lesion.

Features of the lesion, such as location, shape, internal reflectivity, height, and basal diameter, were identified utilizing a 10- or 15-MHz probe. The patients enrolled in the study before 2021 were assessed by a 10-MHz probe, and the rest were examined by a 15-MHz probe due to a renewal in our equipment. The shape was described as a dome, irregular, lobulated, or mushroom. The location of the lesion was noted as posterior pole (P), between the posterior pole and posterior to the equator (PE), posterior to the equator (EP), equator (E), or anterior to the equator (EA). Internal reflectivity was classified as low, low–medium, medium, medium–high, and high.

During the B-scan examination of the lesions, videos were meticulously recorded for subsequent analysis. An experienced echographer (B.A.) thoroughly evaluated these prerecorded videos to assess the intrinsic vascularity of the lesions without the knowledge of the final diagnosis. Areas demonstrating flickering motion within the tumor, which is consistent with the presence of blood flow, were carefully observed. In instances where flickering motion could not be identified, it was noted that intrinsic vascularity was absent. For each tumor, a section with the highest level of detectable flickering motion was selected. Then, the intensity of vascularity was classified according to the proportion of the affected area. Vascularity was categorized as moderate when the flickering motion involved less than half of the cross-sectional area. In contrast, it was classified as intense if the flickering motion occupied half or more of the section's area. We did not evaluate the vascular heterogeneity within the tumor.

Statistical Analysis

Statistical analysis was performed using IBM SPSS Statistics (version 30.0). The variables were investigated using visual (histograms and probability plots) and analytical methods (Kolmogorov–Smirnov test) to determine whether or not they are normally distributed. Quantitative data were expressed as the mean (median, standard deviation, and range), and categorical data as percentages. Pearson chi-square test was used for the categorical data. The Kruskal–Wallis tests were conducted for non-normally distributed and ordinal variables to compare these parameters among 3 groups. The Mann–Whitney *U* test was performed to test the significance of pairwise differences for multiple comparisons. An overall 5% type I error level was used to infer statistical significance. After adjusting the Bonferroni correction, a P value $<$ 0.017 was considered statistically significant. Spearman correlation test assessed correlations between the intensity of

intrinsic vascularity and other parameters. P values <0.05 were considered statistically significant.

Results

Forty-three eyes with CN, 49 eyes with SN, and 45 eyes with CM were included in the study. Figure 1 elicits a demonstration of 3 groups with fundus pictures, FAF, OCT, and B-scan USG. The demographic and clinical features of all groups are listed in Table 1. The median patient age was 71, 64, and 62 years in the CN, SN, and CM groups, respectively. There were no statistically significant differences in sex ($P = 0.39$), involved eye ($P = 0.59$), or the intrinsic pigmentation ($P = 0.29$) of the lesion between the 3 groups, whereas, as shown in Tables 1 and 2, statistically significant differences were observed regarding age, visual acuity, overlying RPE changes, orange pigment on FAF, SRF on OCT, and follow-up time ($P \leq 0.05$). The visual acuity in the CM group was statistically lower than other groups. Since there was no difference in tumor location between groups ($P = 0.19$, Table 3); low visual acuity in the CM group may be related to the presence of extensive SRF involving the macula.

Ultrasonographic features of the lesions were also assessed. As shown in Tables 3 and 4, statistically

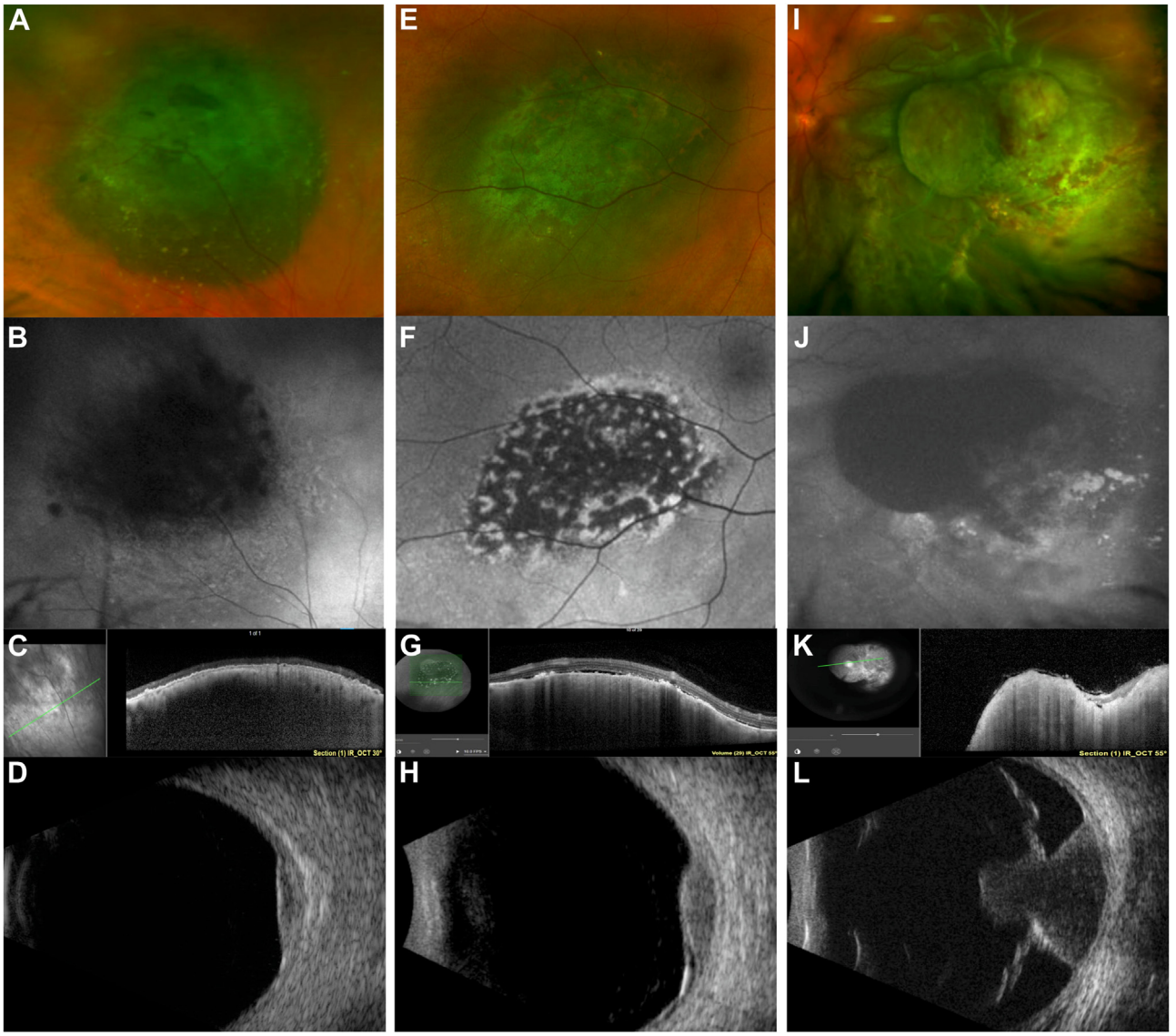


Figure 1. Pigmented choroidal nevus with overlying drusen **(A)**, without orange pigment on FAF **(B)**, without associated subretinal fluid on OCT **(C)**, and medium internal reflectivity on B-scan **(D)**. Pigmented suspicious choroidal nevus with overlying orange pigment **(E)**; orange pigment is more prominent on FAF **(F)**; associated shallow subretinal fluid on OCT **(G)**; low-to-medium internal reflectivity on B-scan **(H)**. Pigmented choroidal melanoma **(I)**, FAF shows areas of hyper- and hypo-autofluorescence **(J)**, OCT demonstrates bilobulated choroidal mass **(K)**, and a mushroom-shaped lesion associated with extensive retinal detachment shows low internal reflectivity on B-scan **(L)**. FAF = fundus autofluorescence.

Table 1. Demographics and Clinical Features of Eyes with Typical Choroidal Nevus, Suspicious Choroidal Nevus, and Choroidal Melanoma at Presentation

Feature	Eye with Typical Choroidal Nevus, n = 43	Eye with Suspicious Choroidal Nevus, n = 49	Eye with Choroidal Melanoma, n = 45	P
Age, mean (median; SD; range), yrs	70.2 (71; 11; 40–94)	63.1 (64; 13; 25–87)	60.3 (62; 15; 22–88)	0.004*
Sex, n (%)				
Female	26 (60)	28 (57)	24 (53)	0.39†
Male	17 (40)	21 (43)	21 (47)	
Involved eye, n (%)				
Right	23 (54)	28 (57)	21 (47)	0.59†
Left	20 (46)	21 (43)	24 (53)	
Visual acuity, mean (median; SD; range), logMAR	0.13 (0; 0.3; −0.2-1.7)	0.14 (0.1; 0.2; 0-0.9)	0.6 (0.3; 0.8; 0-3)	<0.001*
Intrinsic pigmentation, n (%)				
Pigmented	23 (54)	27 (55)	29 (64)	0.29*
Partially pigmented	10 (23)	13 (27)	13 (30)	
Amelanotic	10 (23)	9 (18)	3 (6)	
Overlying RPE changes, n (%)				
Absent	1 (2)	23 (47)	35 (78)	<0.001*
Present	42 (98)	26 (53)	10 (22)	
Orange pigment on FAF, n (%)				
Absent	43 (100)	10 (20)	18 (40)	<0.001*
Present	0 (0)	39 (80)	27 (60)	
Subretinal fluid on OCT, n (%)				
Absent	43 (100)	7 (14)	4 (9)	<0.001*
Present	0 (0)	42 (86)	41 (91)	
Total follow-up time, mean (median; SD; range) mos	71 (66; 28; 24–131)	63 (65; 32; 6–124)	32 (20; 26; 3–78)	<0.001*

FAF = fundus autofluorescence; logMAR = logarithm of the minimum angle of resolution; RPE = retinal pigment epithelium; SD = standard deviation.
*Kruskal–Wallis test.
†Pearson chi-square.

significant differences were observed regarding tumor shape, internal reflectivity, thickness, and basal diameter of the lesion ($P \leq 0.05$). We observed more lobulated and mushroom-shaped lesions in the CM group, whereas we observed more medium–high or high internal reflectivity in the CN group. There was a statistically significant difference in internal reflectivity between CN vs. CM groups ($P < 0.001$) and CN vs. SN groups ($P = 0.014$). Additionally, almost 20% of CN had a low-to-medium internal reflectivity on A-scan reflectivity, and we did not observe any growth during a mean follow-up of 71 months.

Table 2. Post Hoc Tests Regarding Demographics and Clinical Features of Eyes with Typical Choroidal Nevus (Group A), Suspicious Choroidal Nevus (Group B), and Choroidal Melanoma (Group C)

Post Hoc Tests*	A vs. B P	A vs. C P	B vs. C P
Age	0.010	0.002	0.52
LogMAR visual acuity	0.08	<0.001	<0.001
Intrinsic pigmentation	0.74	0.13	0.21
Overlying RPE changes	<0.001	<0.001	0.002
Orange pigment on FAF	<0.001	<0.001	0.03
Subretinal fluid on OCT	<0.001	<0.001	0.41
Total follow-up time	0.35	<0.001	<0.001

A = Eye with typical choroidal nevus group; B = Eye with suspicious choroidal nevus group; C = Eye with choroidal melanoma group; FAF = fundus autofluorescence; logMAR = logarithm of the minimum angle of resolution; RPE = retinal pigment epithelium.
*Mann–Whitney *U* test significance, $P < 0.017$ with Bonferroni correction.

We noted that intrinsic vascularity was observed in 86% of the CN group, 88% of the SN group, and 100% of the CM group. The observation of intrinsic vascularity was significantly different between the 3 groups, and paired post hoc tests showed that the CM group was responsible for the statistical difference. Similarly, we observed the intensity of intrinsic vascularity was predominantly moderate in the CN group (58% of cases) but high in the SN (45% of cases) and CM groups (73% of cases). Furthermore, we investigated the degree of intensity of intrinsic vascularity in the subgroup analysis of the SN group who were observed (n = 33) after initial examination. Twenty-three patients showed no growth within 2 years from the initial USG examination. In this subgroup, 5 (22%) had no detectable intrinsic vascularity, 12 (52%) had moderate, and 6 (26%) had intense intrinsic vascularity at the initial presentation. The remaining 10 patients showed growth of the choroidal lesion within 2 years from the initial USG examination. The degree of intensity of intrinsic vascularity was observed as moderate in 4 patients (40%), high in 5 patients (50%), and not observed in 1 patient (10%) at the initial presentation. Even though the observed degree of intensity of intrinsic vascularity was predominantly intense (50%) in the subgroup with growth within 2 years while moderate (52%) in the subgroup without growth within 2 years, we did not find a statistically significant difference regarding the degree of intensity of vascularity between SN lesions that showed

Table 3. Ultrasonographic Assessment of Eyes with Typical Choroidal Nevus, Suspicious Choroidal Nevus, and Choroidal Melanoma at Presentation

Feature	Eye with Typical Choroidal Nevus, n = 43	Eye with Suspicious Choroidal Nevus, n = 49	Eye with Choroidal Melanoma, n = 45	*P*
Tumor form, n (%)				
Dome	40 (93)	45 (92)	26 (58)	<0.001*
Irregular	1 (2)	2 (4)	0 (0)	
Lobulated	2 (5)	2 (4)	8 (18)	
Mushroom	0 (0)	0 (0)	11 (24)	
Tumor region, n (%)				
Posterior pole	2 (5)	9 (18)	4 (9)	0.19*
Posterior-equator	15 (35)	19 (39)	14 (31)	
Posterior to equator	24 (56)	20 (41)	22 (49)	
Equator	1 (2)	1 (2)	1 (2)	
Anterior to equator	1 (2)	0 (0)	4 (9)	
Internal reflectivity, n (%)				
Low	2 (5)	8 (16)	12 (26)	0.002†
Low–medium	6 (14)	10 (20)	8 (18)	
Medium	17 (40)	20 (41)	17 (38)	
Medium–high	13 (30)	9 (18)	8 (18)	
High	5 (11)	2 (4)	0 (0)	
Tumor height, mean (median; SD; range), mm	1.6 (1.6; 0.3; 1.2–2.3)	1.9 (1.8; 0.4; 1.2–2.9)	6.4 (5.0; 3.4; 1.5–13.8)	<0.001†
Tumor base, mean (median; SD; range), mm	9.69 (9.6; 2.1; 5.5–14.8)	9.1 (8.5; 2.6; 5.9–16.2)	14.9 (13.6; 4.0; 9.7–24.5)	<0.001†
Intrinsic vascularity, n (%)				0.03†
Absent	6 (14)	6 (12)	0 (0)	
Present	37 (86)	43 (88)	45 (100)	
Degree of intensity of intrinsic vascularity, n (%)				
Absent Moderate	6 (14)	6 (12)	0 (0)	<0.001†
Moderate	25 (58)	21 (43)	12 (27)	
Intense	12 (28)	22 (45)	33 (73)	

SD = standard deviation.
*Pearson chi-square test.
†Kruskal–Wallis test.

growth and ones that did not show growth (Mann–Whitney *U* test, $P = 0.017$).

In this cohort, 59 patients were treated with plaque radiotherapy (n = 24 in the SN group and n = 35 in the CM group). In the CM group, 9 patients had primary enucleation, and 3 patients required secondary enucleation after plaque radiotherapy. One patient with CM was followed for only 3 months because the patient had systemic metastasis at the presentation and was lost to follow-up. This study included the results of FNAB for assessment of GEP with DecisionDx-UM (class 1A, class 1B, or class 2) and PRAME RNA expression status with Decision-Dx PRAME (negative or positive) in a total of 64 patients (n = 22 in the SN group and n = 42 in the CM group) to assess the metastatic risk. In the SN group, 13 patients (59%) were in class 1A, 7 patients (32%) in class 1B, and 2 patients (9%) in class 2 group. In the CM group, 18 patients (43%) were in class 1A, 8 patients (19%) in class 1B, and 16 patients (38%) in class 2. Among the patients who had FNAB, PRAME antigen was positive in 5 patients (23%) in the SN group and 16 patients (39%) in the CM group.

Table 4. Post Hoc Tests Regarding Ultrasonographic Assessment of Eyes with Typical Choroidal Nevus (Group A), Suspicious Choroidal Nevus (Group B), and Choroidal Melanoma (Group C)

Post Hoc Tests	A vs. B *P*	A vs. C *P*	B vs. C *P*
Tumor form*	0.83	<0.001	<0.001
Internal reflectivity†	0.014	<0.001	0.28
Tumor height†	0.006	<0.001	<0.001
Tumor base†	0.10	<0.001	<0.001
Presence of intrinsic vascularity†	0.80	0.010	0.016
Degree of intensity of intrinsic vascularity†	0.12	<0.001	0.002

A = Eye with typical choroidal nevus group; B = Eye with suspicious choroidal nevus group; C = Eye with choroidal melanoma group.
*Pearson chi-square test (dome vs. other forms) significance, $P < 0.05$.
†Mann–Whitney *U* test significance, $P < 0.017$ after Bonferroni correction.

We investigated correlations between the observed intensity of intrinsic vascularity and clinical features of the choroidal lesion in the combined group (n = 137) (Table 5). We found a statistically significant positive correlation between the degree of intensity of intrinsic vascularity and the height, basal diameter, and associated SRF ($P < 0.001$, $P < 0.001$, and $P < 0.001$, respectively). In contrast, there was a statistically negative correlation between the degree of intensity of intrinsic vascularity and overlying RPE changes and internal reflectivity ($P = 0.03$ and $P < 0.001$, respectively). Our study could not show a correlation between the degree of intensity of intrinsic

Table 5. Correlations Regarding the Degree of Intensity of Intrinsic Vascularity in Eyes with Choroidal Lesions (Combined Group of Nevus and Melanoma)

	Variable Y	Variable X	n	Spearman Rho	P*
Eye with choroidal lesion (n = 137)	Degree of intensity of intrinsic vascularity	Associated subretinal fluid on OCT	137	0.302	<0.001
		Overlying orange pigment on FAF	137	0.164	0.056
		Overlying RPE changes	137	−0.184	0.03
		Intrinsic pigmentation	137	−0.034	0.69
		Height of the lesion on USG	137	0.470	<0.001
		Base of the lesion on USG	137	0.383	<0.001
		Internal reflectivity on USG	137	−0.407	<0.001
		GEP	64	−0.407	0.80
		PRAME	63	0.149	0.24
		Combined GEP/PRAME status	63	0.085	0.50

FAF = fundus autofluorescence; GEP = gene expression profile (class 1A, class 1B, and class 2 molecular significance); PRAME = preferentially expressed antigen in melanoma (positive or negative); RPE = retinal pigment epithelium; USG = ultrasonography.
Combined GEP/PRAME status in 4 subgroups (class 1/PRAME negative, class 1/PRAME positive, class 2/PRAME negative, and class 2/PRAME positive).
*Spearman correlation test significance, $P < 0.05$.

vascularity and GEP results (class 1A, class 1B, or class 2) ($P = 0.80$), PRAME antigen (positive or negative) ($P = 0.24$), combined GEP/PRAME status (class 1/PRAME-negative, class 1/PRAME-positive, class 2/PRAME-negative, class 2/PRAME-positive) ($P = 0.50$), or orange pigment (present or absent on FAF) ($P = 0.056$). Figure 2 demonstrates the relationship between the lesion's mean height and intrinsic vascularity intensity among the 3 groups. In each group, a higher intensity of intrinsic vascularity was correlated with a larger mean height of the lesion. As expected, key features that are more consistent with CMs, such as larger dimensions, presence of SRF, and low internal reflectivity, are significantly associated with high intensity of intrinsic vascularity (Table 5).

Discussion

As in other tumors, determining blood perfusion level is important to understand the microvascular function and clinical status of the melanocytic uveal tumors. This study aimed to determine whether B-scan USG can detect the intrinsic vascularity of CN and to compare the observed intensity of intrinsic vascularity among typical CN, SN, and CM.

Previous histopathological studies have shown that the cells within CN differ morphologically from normal choroidal melanocytes, appearing atypical yet benign.[8] Rummelt et al used light microscopy and a modified periodic acid–Schiff technique to identify the vascular microcirculatory patterns in CN.[9] They found that nevi exhibited only 4 out of 9 possible vascular patterns:

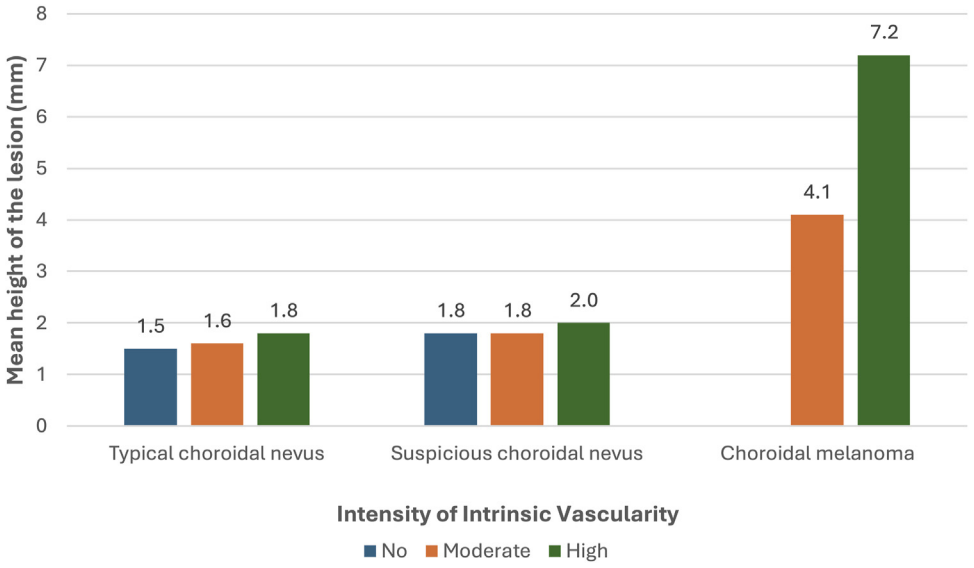


Figure 2. Relationship between the mean height of the lesion and intensity of intrinsic vascularity of eyes with choroidal nevus, suspicious choroidal nevus, and choroidal melanoma.

normal vessels (100%), avascular zones (26%), straight vessels (39%), and parallel straight vessels (17%).[9] In contrast, CM displayed additional vascular patterns, including parallel vessels with cross-linking, arcs, arcs with branching (incomplete loops), microvascular loops encircling small tumor microdomains, and microvascular networks composed of at least 3 consecutive closed loops.[10,11]

These vascular patterns in melanocytic uveal tumors may contribute to the transformation of a CN into melanoma, influencing mortality factors like the ability of neoplastic cells to spread from the eye and evade immune surveillance.[10–13] Ultrastructural studies reveal that the vessels within CN, similar to those in normal choroid, have a thin, homogeneous basement membrane and intact inter-endothelial cell junctions, whereas vessels within CMs have thick, fragmented basement membranes and often lack strong intercellular junctions.[9] These changes are associated with angiogenesis, the formation of new tumor blood vessels.[9]

Pathological intrinsic vascularization of choroidal melanocytic tumors can also be assessed in vivo by indocyanine green angiography.[14] While an avascular (silent) zone has been histologically associated with a favorable diagnosis,[9] Klingenstein et al have shown that the presence of avascular zones on indocyanine green angiography correlates with CN during long-term follow-up.[15] In contrast, parallel vessels with crosslinking and vascular networks were found to be significantly more frequent in CMs.[15]

Previous histopathological studies indicate that the CN may be associated with changes in the overlying tissues, including narrowed choriocapillaris, RPE alterations, drusen formation, and localized retinal detachment.[8] These changes can be evaluated with noninvasive imaging techniques like enhanced-depth imaging OCT[16–20] and OCT angiography.[21–26] Enhanced-depth imaging OCT helps to measure thickness for early detection of growth, especially in flat lesions seen on USG.[19,20] On OCT angiography, nevi exhibit well-defined borders and are hyporeflective,[21–23] with normal overlying retinal and choroidal vasculature.[23–25] Some elevated nevi may show avascular areas[24–26] and an intense vascular rim in the surface microvasculature.[21]

Even though the ultrasonographic assessment may have limitations in detecting flat or minimally elevated CN (<1 mm in height), it is still an indispensable tool for describing acoustic characteristics of choroidal lesions, helping differentiate nevi from melanomas by measuring the baseline tumor thickness and assessing the tumor height growth over time. Traditionally, CN has been typically described as a lesion with high internal reflectivity and lack of intrinsic vascularity on A-scan or B-scan USG.[7] However, advances in ultrasound technology that allow better resolution of anatomic detail can challenge this assumption, as shown in this study. Additionally, previous histopathological studies have shown that CMs typically exhibit a heterogeneous distribution of vascular patterns.[27] However, we were unable to assess this difference in our cohort due to the limitations in ultrasonographic resolution. We speculate that future studies employing Doppler or contrast-enhanced USG or machine-learning algorithms may reveal this heterogeneous distribution more effectively and objectively. In a previous study using Doppler USG, no Doppler signal was elicited from a CN, while pulsatile blood flow was detected at the tumor base of CM.[28] Recent animal studies evaluating the microcirculation of melanocytic uveal tumors with microbubble contrast-enhanced ocular USG demonstrated that contrast-enhanced or VEGF receptor 1–targeted USG could safely detect the microcirculation patterns and quantify blood circulation volume and velocity in animal models with CM.[29–31]

We observed that some CN showing intrinsic vascularity may actually represent small CMs that have not yet entered a phase of rapid growth. Among the patients who were initially monitored due to an SN, 10 demonstrated growth of their lesion within 2 years following the initial ultrasonographic examination. In contrast, 23 patients with SN showed no growth during the same timeframe. However, our study did not reveal a statistically significant difference in the degree of intensity of vascularity between the patients who showed growth or no growth in the SN group. Similarly, we evaluated the degree of intrinsic vascularity between patients within different prognostic groups, including GEP and PRAME status. In our cohort of 64 patients who underwent FNAB to assess GEP and PRAME antigen for evaluating metastatic risk, we found no correlation between the degree of intensity of intrinsic vascularity and either the GEP or PRAME status.

We demonstrated that typical, non-SN can have moderate (58%) or high (28%) intrinsic vascularity, detectable during real-time dynamic B-scan USG examinations (Video S1 [CN] and Video S2 [CM] available at www.ophthalmologyscience.org). This assertion contradicts conventional knowledge, suggesting that CN is a nonvascular lesion in USG.[7] As expected, ultrasound examinations of treatment-naive CMs demonstrated intrinsic vascularity in all cases. We believe that advances in ultrasound technologies and increased experience in ultrasound interpretation have led to higher detection of intrinsic vascularity of CN and melanoma. Therefore, identifying vascular motion on B-scan USG cannot be used to exclude the diagnosis of CN. Serial multimodal imaging along with a clinical follow-up is necessary for the differentiation of CN and melanomas.

Footnotes and Disclosures

Originally received: November 12, 2024.
Final revision: April 23, 2025.
Accepted: April 28, 2025.
Available online: May 6, 2025. Manuscript no. XOPS-D-24-00493.

Department of Ophthalmology and Visual Sciences, Kellogg Eye Center, University of Michigan, Ann Arbor, Michigan.

The material was derived from the American Academy of Ophthalmology Annual Meeting, 2023, poster presentation, November 3-6, San Francisco, California.

Disclosures:

All authors have completed and submitted the ICMJE disclosures form.

The authors made the following disclosures:

H.D.: Consultant – Advisory Board of Castle Biosciences, Advisory Board of Immunocore.

This study was partly supported by the Richard N. and Marilyn K. Witham Professorship (H.D.). The founder has no effect on the design, result, and writing of the manuscript.

Support for Open Access publication was provided by the University of Michigan.

HUMAN SUBJECTS: Human subjects were included in this study. The study adhered to the tenets of the Declaration of Helsinki. The review of the medical records and ultrasonographic examination was conducted under the study approved by the Institutional Review Board of the University of Michigan. Therefore, the informed consent for each patient was not applicable.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Beser, McClendon, Ayres, Demirci

Data collection: Beser, McClendon, Ayres, Demirci

Analysis and interpretation: Beser, McClendon, Ayres, Demirci

Obtained funding: N/A

Overall responsibility: Beser, McClendon, Ayres, Demirci

Abbreviations and Acronyms:
CM = choroidal melanoma; **CN** = choroidal nevi; **FAF** = fundus autofluorescence; **FNAB** = fine needle aspiration biopsy; **GEP** = gene expression profile; **PRAME** = preferentially expressed antigen in melanoma; **RPE** = retinal pigment epithelium; **SN** = suspicious choroidal nevi; **SRF** = subretinal fluid; **USG** = ultrasonography.

Keywords:
Choroidal nevus, Suspicious choroidal nevus, Choroidal melanoma, B-scan ultrasonography, Intrinsic vascularity.

Correspondence:
Hakan Demirci, MD, Department of Ophthalmology and Visual Sciences, Kellogg Eye Center, University of Michigan, 1000 Wall St, Rm 641, Ann Arbor, MI 48105. E-mail: hdemirci@med.umich.edu.

References

1. Shields JA, Shields CL. Choroidal nevus. In: *Intraocular tumors. An atlas and textbook*. 3rd ed. Philadelphia, PA: Lippincott Williams and Wikins, Wolters Kluwers; 2015: 69–80.
2. Qiu M, Shields CL. Choroidal nevus in the United States adult population: racial disparities and associated factors in the national health and nutrition examination survey. *Ophthalmology*. 2015;122:2071–2083.
3. Chien JL, Sioufi K, Surakiatchanukul T, et al. Choroidal nevus: a review of prevalence, features, genetics, risks, and outcomes. *Curr Opin Ophthalmol*. 2017;28:228–237.
4. Li HK, Shields CL, Mashayekhi A, et al. Giant choroidal nevus clinical features and natural course in 322 cases. *Ophthalmology*. 2010;117:324–333.
5. Verbeek S, Dalvin LA. Advances in multimodal imaging for diagnosis of pigmented ocular fundus lesions. *Can J Ophthalmol*. 2024;59:218–233.
6. Collaborative Ocular Melanoma Study Group, Boldt HC, Byrne SF. Baseline echographic characteristics of tumors in eyes of patients enrolled in the Collaborative Ocular Melanoma Study: COMS report no. 29. *Ophthalmology*. 2008;115: 1390–1397.e2.
7. Byrne SF, Green GL. Intraocular tumors. In: Byrne SF, ed. *Ultrasound of the Eye and Orbit*. 2nd ed. St. Louis: Mosby; 2002:115–190.
8. Naumann G, Yanoff M, Zimmerman LE. Histogenesis of malignant melanomas of the uvea. I. Histopathologic characteristics of nevi of the choroid and ciliary body. *Arch Ophthalmol*. 1966;76:784–796.
9. Rummelt V, Folberg R, Rummelt C, et al. Microcirculation architecture of melanocytic nevi and malignant melanomas of the ciliary body and choroid. A comparative histopathologic and ultrastructural study. *Ophthalmology*. 1994;101: 718–727.
10. Folberg R, Rummelt V, Parys-Van Ginderdeuren R, et al. The prognostic value of tumor blood vessel morphology in primary uveal melanoma. *Ophthalmology*. 1993;100: 1389–1398.
11. Pe'er J, Rummelt V, Mawn L, et al. Mean of the ten largest nucleoli, microcirculation architecture, and prognosis of ciliochoroidal melanomas. *Ophthalmology*. 1994;101: 1227–1235.
12. Mueller AJ, Bartsch DU, Folberg R, et al. Imaging the microvasculature of choroidal melanomas with confocal indocyanine green scanning laser ophthalmoscopy. *Arch Ophthalmol*. 1998;116:31–39.
13. Kivelä T, Mäkitie T, Al-Jamal RT, Toivonen P. Microvascular loops and networks in uveal melanoma. *Can J Ophthalmol*. 2004;39:409–421.
14. Kubicka-Trzaska A, Starzycka M, Romanowska-Dixon B. Indocyanine green angiography in the diagnosis of small choroidal tumours. *Ophthalmologica*. 2002;216:316–319.
15. Klingenstein A, Schaumberger MM, Freeman WR, et al. MuSIC report III: tumour microcirculation patterns and development of metastasis in long-term follow-up of melanocytic uveal tumours. *Acta Ophthalmol*. 2016;94:175–181.
16. Torres VL, Brugnoni N, Kaiser PK, Singh AD. Optical coherence tomography enhanced depth imaging of choroidal tumors. *Am J Ophthalmol*. 2011;151:586–593.e2.
17. Shah SU, Kaliki S, Shields CL, et al. Enhanced depth imaging optical coherence tomography of choroidal nevus in 104 cases. *Ophthalmology*. 2012;119:1066–1072.
18. Shields CL, Manalac J, Das C, et al. Review of spectral domain enhanced depth imaging optical coherence tomography of tumors of the choroid. *Indian J Ophthalmol*. 2015;63: 117–121.
19. Jonna G, Daniels AB. Enhanced depth imaging OCT of ultrasonographically flat choroidal nevi demonstrates 5 distinct patterns. *Ophthalmol Retina*. 2019;3:270–277.
20. Martins MF, Kiefer K, Kanecadan LAA, et al. Comparisons of choroidal nevus measurements obtained using 10- and 20-MHz ultrasound and spectral domain optical coherence tomography. *Arq Bras Oftalmol*. 2017;80:78–83.
21. Ghassemi F, Mirshahi R, Fadakar K, Sabour S. Optical coherence tomography angiography in choroidal melanoma and nevus. *Clin Ophthalmol*. 2018;12:207–214.

22. Gündüz AK, Mirzayev I, Kasimoglu R, Özalp Ateş FS. Swept-source optical coherence tomography angiography findings in choroidal and retinal tumors. *Eye (Lond)*. 2021;35:4−16.
23. Garcia-Arumi Fuste C, Peralta Iturburu F, Garcia-Arumi J. Is optical coherence tomography angiography helpful in the differential diagnosis of choroidal nevus vs. melanoma? *Eur J Ophthalmol*. 2020;30:723−729.
24. Greig EC, Laver NV, Mendonca LSM, et al. SWEPT-SOURCE optical coherence tomography angiography in small choroidal melanomas and choroidal nevi. *Retina*. 2021;41:1182−1192.
25. Cennamo G, Romano MR, Breve MA, et al. Evaluation of choroidal tumors with optical coherence tomography: enhanced depth imaging and OCT-angiography features. *Eye (Lond)*. 2017;31:906−915.
26. Toledo JJ, Asencio-Duran M, García-Martinez JR, López-Gaona A. Use of OCT angiography in choroidal melanocytic tumors. *J Ophthalmol*. 2017;2017:1573154.
27. Folberg R, Pe'er J, Gruman LM, et al. The morphologic characteristics of tumor blood vessels as a marker of tumor progression in primary human uveal melanoma: a matched case-control study. *Hum Pathol*. 1992;23:1298−1305.
28. Wolff-Korman PG, Kormann BA, Hasenfratz GC, Spengel FA. Duplex and color Doppler ultrasound in the differential diagnosis of choroidal tumors. *Acta Ophthalmol Suppl (1985)*. 1992:66−70.
29. Zhang Q, Yang H, Kang SJ, et al. In vivo high-frequency, contrast-enhanced ultrasonography of uveal melanoma in mice: imaging features and histopathologic correlations. *Invest Ophthalmol Vis Sci*. 2011;52:2662−2668.
30. Gao M, Tang J, Liu K, et al. Quantitative evaluation of vascular microcirculation using contrast-enhanced ultrasound imaging in rabbit models of choroidal melanoma. *Invest Ophthalmol Vis Sci*. 2018;59:1251−1262.
31. Liu H, Gao M, Gu J, et al. VEGFR1-Targeted contrast-enhanced ultrasound imaging quantification of vasculogenic mimicry microcirculation in a mouse model of choroidal melanoma. *Transl Vis Sci Technol*. 2020;9:4.